

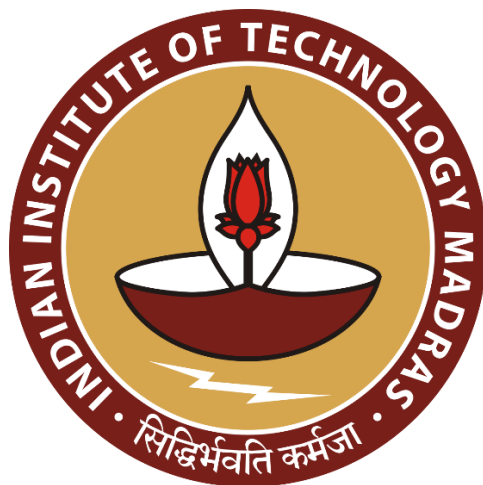
Multimodal Misinformation Detector Agent

Data Science and AI Lab Project

Milestone 1 Report

Group 9

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1. Abstract

The rise of synthetic media, such as Deep Fakes and AI-generated text, presents major risks to information integrity and digital trust. Malicious actors exploit these tools to spread disinformation, manipulate public perception, and compromise individuals' reputations. This project proposes a two-part system to address this challenge.

A vision-based classifier detects whether an uploaded image is real or manipulated using fine-tuned image model ***openai/clip-vit-base-patch32***. A text credibility pipeline evaluates user-submitted text by retrieving evidences from Google Fact Check, Tavily web-search API, and Wikimedia sources. The model ***microsoft/deberta-v3-base*** fine-tuned on the FEVER dataset classifies the claim as real or fake based on the gathered evidence, while ***mistralai/Mistral-7B-Instruct-v0.2*** produces a natural-language explanation of the final credibility assessment.

2. Objectives

- **Deep Fake Detection** – Develop an image-based classifier capable of identifying synthetic images by using deep learning models trained on labeled datasets of real and fake visuals. The system should generalize well across diverse sources, lighting conditions, and manipulation techniques.
- **Text Credibility Analysis** – Enable users to submit text and receive a detailed credibility report through Natural Language Processing and fact-checking algorithms.
- **End-to-End User Experience** – Provide a seamless and intuitive interface for users to submit both images and text, ensuring rapid and transparent feedback. The platform will emphasize usability, accessibility, and interpretability of results.

This work primarily benefits journalists combating misinformation, educators teaching media literacy, fact-checkers, and the general public who want quick verification tools.

3. Literature Review

3.1. Existing Approaches in Image Deep Fake Detection

Deepfake detection has advanced with large-scale datasets and deep learning. **FaceForensics++** introduced over 1.8 million manipulated images, enabling standardized evaluation; **XceptionNet** achieved strong detection under compression (Rössler et al., 2019). The DeepFake Detection Challenge (DFDC) released >100,000 videos, where ensemble CNNs achieved $AUC \approx 0.90$, but performance declined on unseen manipulations (Dolhansky et al., 2020).

Lightweight CNNs such as **MesoNet** (Afchar et al., 2018) and forensic methods analyzing **GAN fingerprints** (Marra et al., 2019) focus on efficiency and artifact detection. However, most models rely solely on visual cues, limiting cross-domain robustness. Recent work suggests that vision–language models like **CLIP** (Radford et al., 2021) could enhance generalization by leveraging semantic alignment between image and text.

3.2 Existing Approaches in Text Credibility and Fact-Checking

Automated fact-checking typically follows a **Retrieve–Rerank–Verify** pipeline. The **FEVER** dataset contains 185,445 claims linked to Wikipedia evidence; baseline systems achieved 31.9% accuracy, and best shared-task systems reached a **FEVER score of ~64%** (Thorne et al., 2018). Later datasets such as **FEVEROUS** (Aly et al., 2021) added tabular evidence to test multi-modal reasoning.

Modern systems use Transformer-based models like **BERT** and **DeBERTa** for claim verification (Zhou et al., 2019). Surveys highlight two open challenges—low retrieval recall and lack of explainability (Kotonya & Toni, 2020; Guo et al., 2022). **RAG** methods (Lewis et al., 2020) now combine external evidence with large language models to generate explanations, though evidence freshness and grounding remain issues.

3.3 Contribution of Proposed Solution

This work proposes an integrated multimodal credibility pipeline addressing both visual and textual misinformation:

- A **CLIP-ViT-B/32** image classifier detects manipulated images, extending beyond CNN-only detectors.
- Text claims are verified using **Google Fact Check**, **Tavily**, and **Wikimedia** retrieval, enhancing evidence diversity
- A **DeBERTa-v3-base** model fine-tuned on FEVER predicts real or fake based on retrieved evidence.
- **Mistral-7B-Instruct** generates human-readable explanations for interpretability.

Together, these components bridge gaps in prior work by combining multi-source evidence retrieval, vision–language reasoning, and explainable decision-making.

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4. Model Architecture

4.1 Deep Fake Detection (Image-Only Submission)

- **Input:** User-submitted image.
- **Model:** Pre trained Vision Model - CLIP-ViT-B/32
 - **Finetuning:** The model will be finetuned on a carefully curated dataset of "real" and "fake" images, with a focus on diverse deepfake generation methods.
 - **Output Layer:** A classification head (e.g., a fully connected layer with a sigmoid activation) will output a probability score indicating whether the image is a fake or not (e.g., 0 for real, 1 for fake).
- **Output:** Binary classification (Fake/Real).

4.2 Disinformation Detection (Text-Only Submission)

- **Input:** User-submitted text.
- **Component 1: Web Evidence Retrieval:**
 - **Function:** Uses APIs (e.g., **Tavily**, **Google Fact Check**, **Wikimedia**) to perform targeted web searches for each summarized claim
 - **Output:** A curated set of evidence — including links, snippets, and factual excerpts — from diverse and credible online sources.
- **Component 2: Credibility Report Generation:**
 - **Input:** Original text, summarized claims, and retrieved evidences.
 - **Function:**
 - **DeBERTa-v3** evaluates factual accuracy.

- **Mistral-7B** composes a natural-language explanation.
- The system assesses source reliability, bias indicators, and emotional tone.
- **Output:** A detailed credibility report, potentially including a numerical score or qualitative assessment (e.g., "Highly Credible," "Partially True," "Misleading," "False").

5. Dataset

To support the image forgery detection pipeline, we will rely on publicly available labeled image datasets that provide clear ground truth (real vs fake). For finetuning DeBERTa-v3, we use FEVER dataset from Huggingface.

- **Kaggle:** [Kaggle – AI-Generated Images vs Real Images](#)

This dataset presents a balanced and diverse collection of 60,000 labeled images, equally divided between AI-generated and real categories, ensuring fair representation for supervised model training. The data spans a wide range of visual domains including landscapes, people, objects, and abstract art, enabling the model to generalize beyond face-centric or narrow-domain forgeries..

- **FEVER:** [HuggingFace – FEVER Dataset](#)

The **FEVER (Fact Extraction and VERification)** dataset, released by the University of Sheffield and the University of Washington, is a large-scale benchmark designed for automated fact verification tasks. It contains approximately **185,445** human-generated claims, each annotated as SUPPORTED, REFUTED, or NOT ENOUGH INFO, based on evidence retrieved from Wikipedia. For each claim, relevant evidence sentences are also provided, enabling both classification and retrieval-based evaluation.

- **Web Evidence (for Text Credibility Pipeline)**

For text credibility assessment, we will not rely on a static dataset but instead use real-time search APIs. These APIs will retrieve external evidence, which the LLM will use to support or refute the input claims.

6. Feasibility and Compute

- **Compute Resources:** GPU-backed cloud instances (e.g., Google Colab Pro or AWS EC2). CLIP fine-tuning requires moderate resources (1–2 GPUs). LLM access will be via API.
- **Implementation Feasibility:** Leveraging existing pretrained models and datasets reduces training time. The main challenge is integration and evaluation.
- **Ethical Considerations:** The tool will include disclaimers to prevent misuse, avoid overconfidence in model predictions, and ensure transparency.

7. Evaluation Metrics

- **Image Detection:** Accuracy, F1-score, ROC-AUC on benchmark datasets.
- **Text Credibility:** Alignment with fact-checking sources, human evaluation of reports, consistency across queries.
- **User Experience:** Usability feedback and response time.

8. Roles

- **Project Manager** – Coordination, documentation, milestone tracking.
- **Vision Lead** – Dataset preparation, fine-tuning CLIP, image evaluation pipeline.
- **NLP Lead** – LLM summarization, credibility report design.
- **Backend/Integration Lead** – Connecting pipelines with APIs, system architecture.
- **Frontend/UI Lead** – Interface design and user experience.

9. Timeline

- **Week 1-2:** Literature review, dataset curation, initial UI sketches.
- **Week 3-4:** Image pipeline setup, preliminary CLIP fine-tuning.
- **Week 5-6:** Text pipeline setup with Gemini API + web search integration.
- **Week 7-8:** Integration testing + Evaluation +Final report.

10. Conclusion

This milestone establishes the foundation for a unified system tackling misinformation in both visual and textual domains. By fine-tuning multimodal models for DeepFake detection and coupling LLM summarization with evidence retrieval for text credibility, the system addresses a pressing societal need. The team will continue to refine datasets, pipelines, and user interfaces in subsequent milestones.